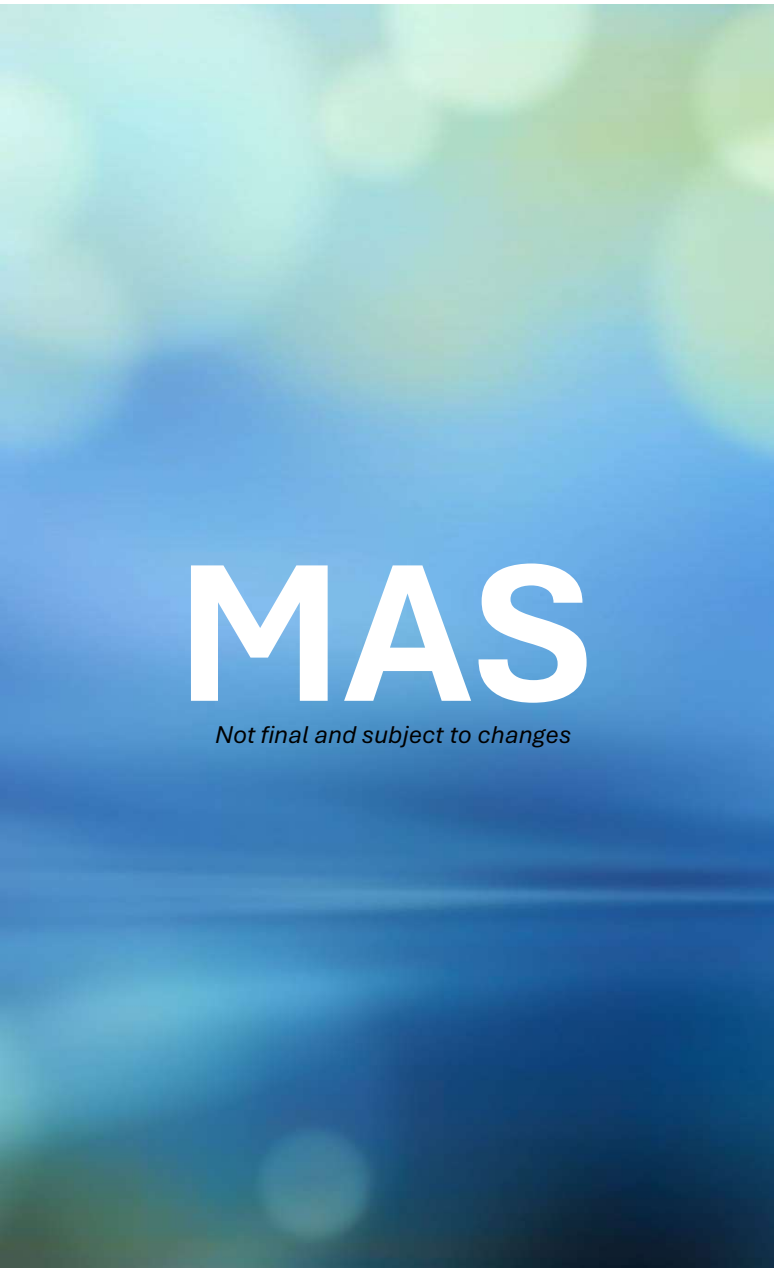
Mechanical **A**erial **S**ystems
(MAS)

F26, SDU Dronecenter

Robert Ladig – rola@sdu.dk



Not final and subject to changes

Unit 1 – Course goal, Intro, Housekeeping, Aerial Manipulation Intro, Simulator Setup, ROS2 (refresher ?!)
Unit 2 - Manipulator types, FK (refresher), IK (refresher), URDF (refresher?!), Simulator work

Unit 3 – Jacobians and coupled kinematics, Simulator work
Unit 4 – PID control (refresher!), Simulator implementation of coupled kinematics for SDU UAS system
Unit 5 – Material handover, Material check and prototyping of own aerial manipulator.

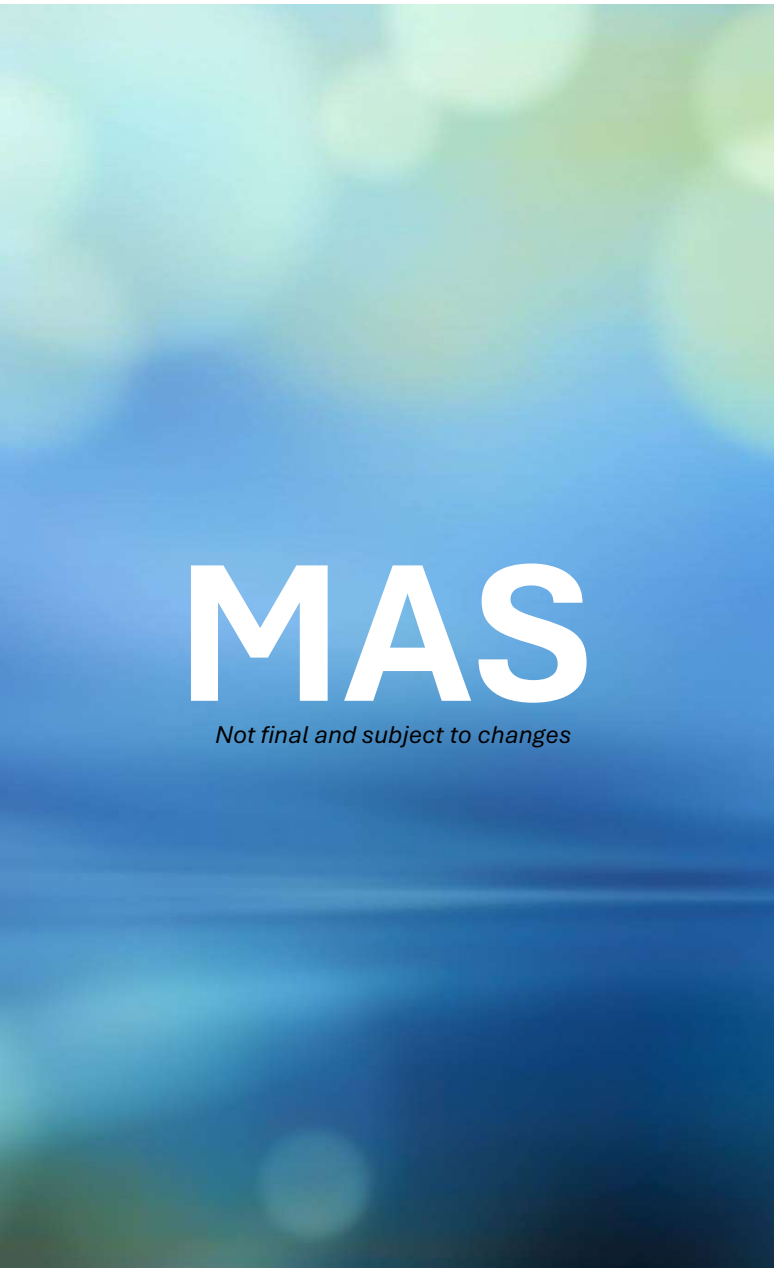
Unit 6 - Group presentation of concept, CAD and refined Unit 4 simulator

----------Flight lab is available to use and drones are RTF----------

Unit 7 – Intro to Flightlab, EIT playground, Offboard control test @ Flightlab
Unit 8 - Bench prototype of Aerial Manipulator
Unit 9 - Airframe integration
Unit 10 – Airframe integration, Flight test I, (opt.) Sensing and control integration

Unit 11 – Final group presentation, Flight test II
Unit 12 – Flight demo , Project report due

Not final and subject to changes



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Unit 1 – Course goal, Intro, Housekeeping, Aerial Manipulation Intro, Simulator Setup
Unit 2 - Manipulator types, FK (refresher), ROS2 (refresher ?!), URDF (refresher?!), Simulator work

Unit 3 – IK (refresher), Jacobians, Simulator work
Unit 4 – Coupled kinematics, Nullspace, ~~PID control (refresher!)~~, Simulator implementation of coupled kinematics for SDU UAS system
Unit 5 – Git etiquette, Material handover, Material check and prototyping of own aerial manipulator.

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$$\dot{\mathbf{p}} = J(\boldsymbol{\theta})\,\dot{\boldsymbol{\theta}}, \qquad \dot{\boldsymbol{\theta}} = J(\boldsymbol{\theta})^{+}\,\dot{\mathbf{p}}_{\mathrm{des}}$$

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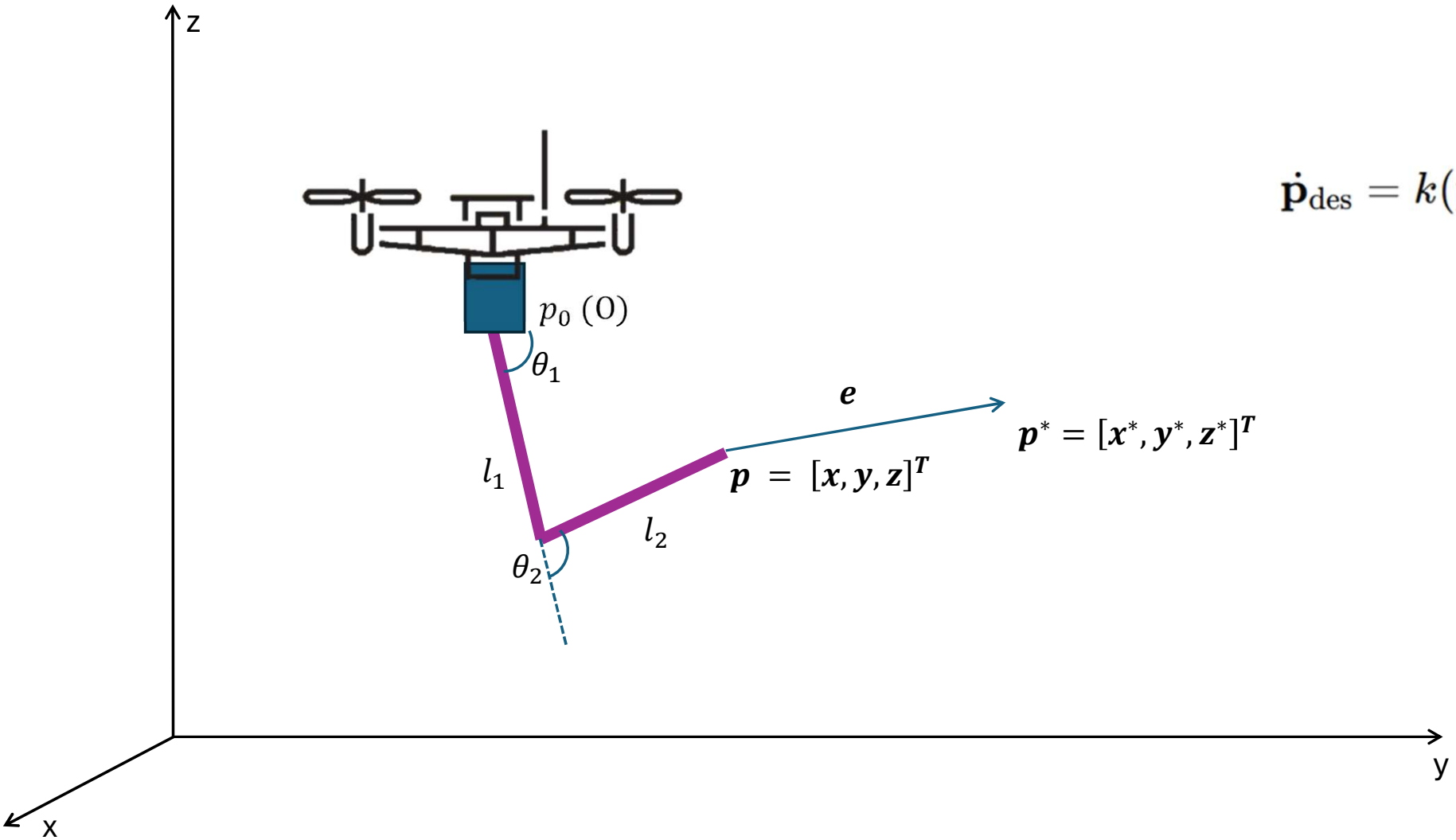
$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} J_{\text{base}}(\cdot) & J_{\text{arm}}(\boldsymbol{\theta}) \end{bmatrix}}_{J_{\text{mobile}}} \begin{bmatrix} \dot{\mathbf{q}}_{\text{base}} \\ \dot{\boldsymbol{\theta}} \end{bmatrix}$$

$$\dot{\mathbf{q}}_{\text{base}} = \overset{\text{planar}}{[v_x, v_y, \omega_z]^T} \qquad \overset{\text{spatial}}{[v_x, v_y, v_z, \omega_x, \omega_y, \omega_z]^T}$$



$$\dot{\mathbf{p}}_{\text{des}} = k(\mathbf{p}^* - \mathbf{p})$$

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} J_{\text{base}}(\cdot) & J_{\text{arm}}(\boldsymbol{\theta}) \end{bmatrix}}_{J_{\text{mobile}}} \begin{bmatrix} \dot{\mathbf{q}}_{\text{base}} \\ \dot{\boldsymbol{\theta}} \end{bmatrix}$$

World position of End-Effector

$$\mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}, \qquad \dot{\mathbf{p}} = \begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \end{bmatrix}$$

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} J_{\text{base}}(\cdot) & J_{\text{arm}}(\boldsymbol{\theta}) \end{bmatrix}}_{J_{\text{mobile}}} \begin{bmatrix} \dot{\mathbf{q}}_{\text{base}} \\ \dot{\boldsymbol{\theta}} \end{bmatrix}$$

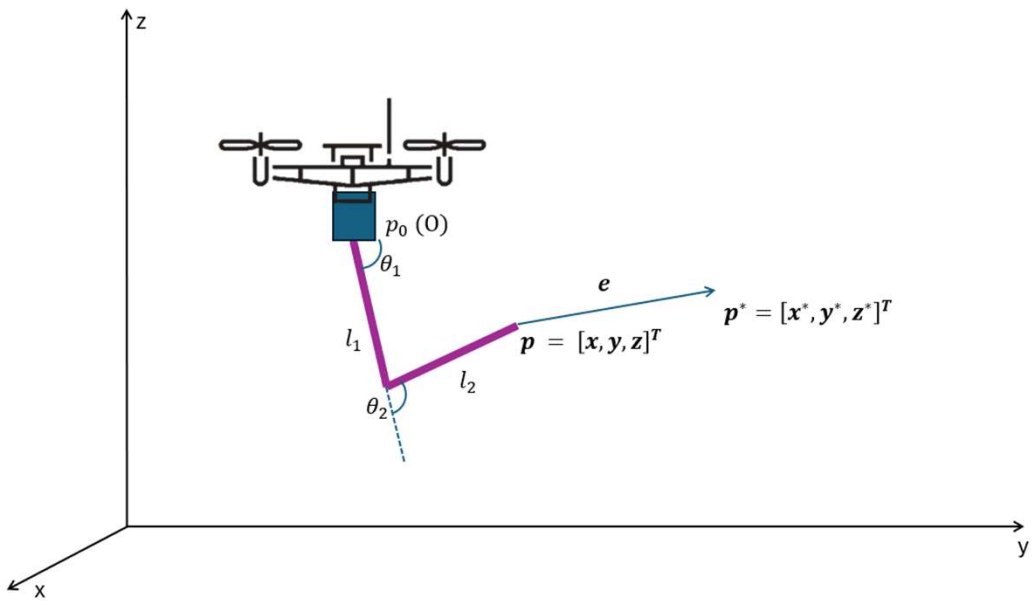
High level drone base state

$$\mathbf{q}_{\text{base}} = \begin{bmatrix} x \\ y \\ z \\ \psi \end{bmatrix}, \qquad \dot{\mathbf{q}}_{\text{base}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\psi} \end{bmatrix}$$

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} J_{\text{base}}(\cdot) & J_{\text{arm}}(\boldsymbol{\theta}) \end{bmatrix}}_{J_{\text{mobile}}} \begin{bmatrix} \dot{\mathbf{q}}_{\text{base}} \\ \dot{\boldsymbol{\theta}} \end{bmatrix}$$

Arm joints

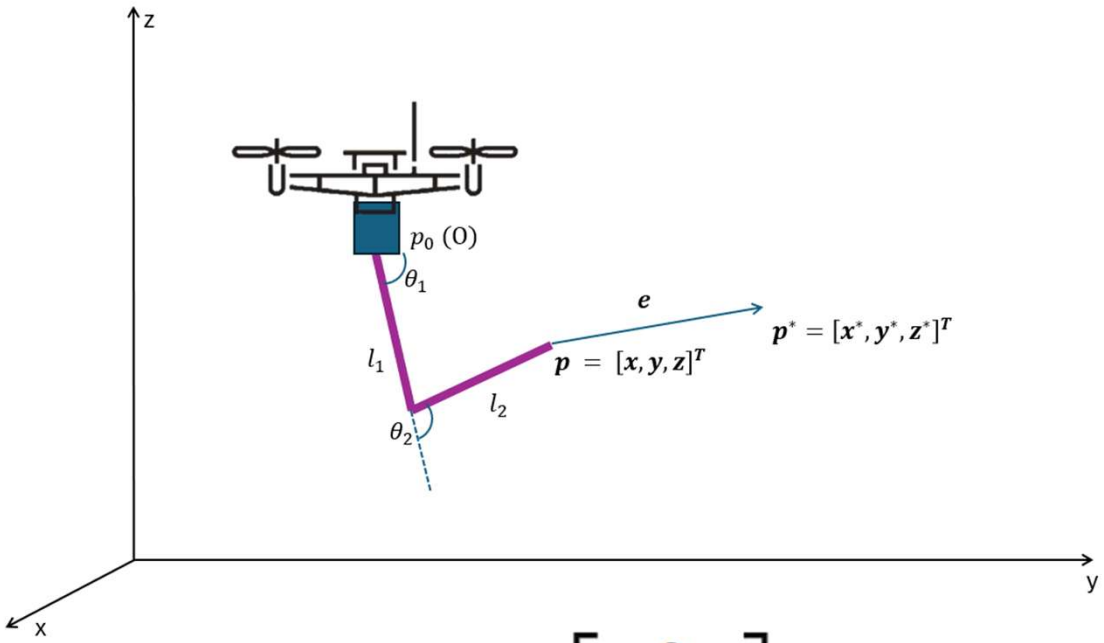
$$\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \qquad \dot{\theta} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$



$$\mathbf{p} = \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbf{p}_{\text{base}}} + \underbrace{R_z(\psi)}_{\text{yaw}} \underbrace{\mathbf{r}(\theta)}_{\text{arm tip in gimbal frame}}$$

$$\mathbf{p} = \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbf{p}_{\text{base}}} + \underbrace{R_z(\psi)}_{\text{yaw}} \underbrace{\mathbf{r}(\theta)}_{\text{arm tip in gimbal frame}}$$

$$R_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$\mathbf{r}(\theta) = \begin{bmatrix} 0 \\ r_y(\theta) \\ r_z(\theta) \end{bmatrix}$$

$$r_y(\theta) = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$r_z(\theta) = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} J_{\text{base}}(\cdot) & J_{\text{arm}}(\boldsymbol{\theta}) \end{bmatrix}}_{J_{\text{mobile}}} \begin{bmatrix} \dot{\mathbf{q}}_{\text{base}} \\ \dot{\boldsymbol{\theta}} \end{bmatrix}$$

$$\mathbf{p} = \mathbf{p}_{\text{base}} + R_z(\psi)\mathbf{r}(\theta)$$

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}}_{\text{base translation}} + \underbrace{\frac{dR_z(\psi)}{d\psi}\,\mathbf{r}(\theta)\,\dot{\psi}}_{\text{yaw term}} + \underbrace{R_z(\psi)\,\dot{\mathbf{r}}(\theta)}_{\text{arm term}}$$

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}}_{\text{base translation}} + \underbrace{\frac{dR_z(\psi)}{d\psi}\,\mathbf{r}(\theta)\,\dot{\psi}}_{\text{yaw term}} + \underbrace{R_z(\psi)\,\dot{\mathbf{r}}(\theta)}_{\text{arm term}}$$

$$\mathbf{r}(\theta) = \begin{bmatrix} 0 \\ r_y(\theta) \\ r_z(\theta) \end{bmatrix}, \quad r_y(\theta) = l_1\cos(\theta_1) + l_2\cos(\theta_1+\theta_2), \quad r_z(\theta) = l_1\sin(\theta_1) + l_2\sin(\theta_1+\theta_2)$$

$$\dot{\mathbf{r}}(\theta) = J_{\text{arm}}^{(G)}(\theta)\,\dot{\theta}$$

$$J_{\text{arm}}^{(G)}(\theta) = \begin{bmatrix} 0 & 0 \\ -l_1\sin(\theta_1) - l_2\sin(\theta_1+\theta_2) & -l_2\sin(\theta_1+\theta_2) \\ l_1\cos(\theta_1) + l_2\cos(\theta_1+\theta_2) & l_2\cos(\theta_1+\theta_2) \end{bmatrix}$$

$$R_z(\psi)\,\dot{\mathbf{r}}(\theta) = \underbrace{R_z(\psi)\,J_{\text{arm}}^{(G)}(\theta)}_{J_{\text{arm}}(\psi,\theta)}\dot{\theta}$$

$$a_1 = -l_1 \sin(\theta_1) - l_2 \sin(\theta_1 + \theta_2), \qquad a_2 = -l_2 \sin(\theta_1 + \theta_2)$$

$$b_1 = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2), \qquad b_2 = l_2 \cos(\theta_1 + \theta_2)$$

$$\boxed{J_{\text{arm}}(\psi, \theta) = \begin{bmatrix} -\sin\psi\, a_1 & -\sin\psi\, a_2 \\ \cos\psi\, a_1 & \cos\psi\, a_2 \\ b_1 & b_2 \end{bmatrix}}$$

(Again)

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}}_{\text{base translation}} + \underbrace{\frac{dR_z(\psi)}{d\psi}\,\mathbf{r}(\theta)\,\dot{\psi}}_{\text{yaw term}} + \cancel{\underbrace{R_z(\psi)\,\dot{\mathbf{r}}(\theta)}_{\text{arm term}}}$$

$$R_z(\psi)\mathbf{r}(\theta) = \begin{bmatrix} -\sin\psi\, r_y(\theta) \\ \cos\psi\, r_y(\theta) \\ r_z(\theta) \end{bmatrix}$$

$$\frac{dR_z(\psi)}{d\psi}\,\mathbf{r}(\theta) = \begin{bmatrix} -\cos\psi\, r_y(\theta) \\ -\sin\psi\, r_y(\theta) \\ 0 \end{bmatrix}$$

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & -\cos\psi\, r_y(\theta) \\ 0 & 1 & 0 & -\sin\psi\, r_y(\theta) \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{J_{\text{base}}(\psi,\theta)} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\psi} \end{bmatrix} + J_{\text{arm}}(\psi,\theta)\,\dot{\theta}$$

Lets bring it all together!

Stack all controllable rates

$$\dot{\mathbf{q}} = \begin{bmatrix} \dot{\mathbf{q}}_{\text{base}} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\psi} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

Lets bring it all together!

End- effector velocity

$$\dot{\mathbf{p}} = J_{\text{mobile}}(\psi, \theta)\, \dot{\mathbf{q}}$$

with

$$J_{\text{mobile}}(\psi, \theta) = \left[J_{\text{base}}(\psi, \theta) \;\; J_{\text{arm}}(\psi, \theta) \right]$$

Lets bring it all together!

$$\dot{\mathbf{p}} = \begin{bmatrix} J_{\text{base}}(\psi, \theta) & J_{\text{arm}}(\psi, \theta) \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\psi} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$$\dot{\mathbf{p}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & -\cos\psi\, r_y(\theta) \\ 0 & 1 & 0 & -\sin\psi\, r_y(\theta) \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{J_{\text{base}}(\psi,\theta)} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\psi} \end{bmatrix} + J_{\text{arm}}(\psi,\theta)\,\dot{\theta}$$

$$R_z(\psi)\,\dot{\mathbf{r}}(\theta) = \underbrace{R_z(\psi)\,J_{\text{arm}}^{(G)}(\theta)}_{J_{\text{arm}}(\psi,\theta)}\dot{\theta}$$

$$a_1 = -l_1 \sin(\theta_1) - l_2 \sin(\theta_1 + \theta_2), \qquad a_2 = -l_2 \sin(\theta_1 + \theta_2)$$

$$b_1 = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2), \qquad b_2 = l_2 \cos(\theta_1 + \theta_2)$$

$$\boxed{J_{\text{arm}}(\psi, \theta) = \begin{bmatrix} -\sin\psi\, a_1 & -\sin\psi\, a_2 \\ \cos\psi\, a_1 & \cos\psi\, a_2 \\ b_1 & b_2 \end{bmatrix}}$$

$$a_1 = -l_1 \sin(\theta_1) - l_2 \sin(\theta_1 + \theta_2), \quad a_2 = -l_2 \sin(\theta_1 + \theta_2)$$

$$b_1 = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2), \quad b_2 = l_2 \cos(\theta_1 + \theta_2)$$

$$r_y(\theta) = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$\boxed{J_{\text{mobile}}(\psi, \theta) = \begin{bmatrix} 1 & 0 & 0 & -\cos\psi\, r_y(\theta) & -\sin\psi\, a_1 & -\sin\psi\, a_2 \\ 0 & 1 & 0 & -\sin\psi\, r_y(\theta) & \cos\psi\, a_1 & \cos\psi\, a_2 \\ 0 & 0 & 1 & 0 & b_1 & b_2 \end{bmatrix}}$$

Finally:

$$\mathbf{p}^* = \begin{bmatrix} p_x^* \\ p_y^* \\ p_z^* \end{bmatrix}, \qquad \mathbf{e} = \mathbf{p}^* - \mathbf{p}$$

$$\dot{\mathbf{p}}_{\text{des}} = k\,\mathbf{e}$$

$$\dot{\mathbf{q}} = J_{\text{mobile}}(\psi, \theta)^+ \, \dot{\mathbf{p}}_{\text{des}}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\psi} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = J_{\text{mobile}}(\psi, \theta)^+ \begin{bmatrix} \dot{p}_{x,\text{des}} \\ \dot{p}_{y,\text{des}} \\ \dot{p}_{z,\text{des}} \end{bmatrix}$$

Numerical example:

$$\mathbf{p}_{\text{base}} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \quad \psi = 30^\circ, \quad \theta_1 = -70^\circ,\ \theta_2 = 100^\circ, \quad l_1 = 0.30,\ l_2 = 0.20$$

$$\mathbf{p}^* = \begin{bmatrix} 2 \\ 3 \\ 1.5 \end{bmatrix}$$

$$\mathbf{r}(\theta) = \begin{bmatrix} 0 \\ r_y(\theta) \\ r_z(\theta) \end{bmatrix}$$

$$r_y = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) = 0.2758$$

$$r_z = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2) = -0.1819$$

with

$$R_z(\psi)[0, r_y, r_z]^T = [-\sin\psi\, r_y,\ \cos\psi\, r_y,\ r_z]^T$$

$$\mathbf{p} = \mathbf{p}_{\text{base}} + \begin{bmatrix} -\sin\psi\, r_y \\ \cos\psi\, r_y \\ r_z \end{bmatrix} = \begin{bmatrix} -0.1379 \\ 1.2389 \\ 1.8181 \end{bmatrix}$$

$$\text{choose } k = 0.2\,\mathrm{s}^{-1}$$

$$\mathbf{e} = \mathbf{p}^* - \mathbf{p} = \begin{bmatrix} 2.1379 \\ 1.7611 \\ -0.3181 \end{bmatrix}$$

$$\dot{\mathbf{p}}_{\mathrm{des}} = k\mathbf{e} = \begin{bmatrix} 0.4276 \\ 0.3522 \\ -0.0636 \end{bmatrix} \ \mathrm{m/s}$$

$$J_{\text{mobile}} = [J_{\text{base}} \;\; J_{\text{arm}}]$$

$$J_{\text{base}} = \begin{bmatrix} 1 & 0 & 0 & -\cos\psi\, r_y \\ 0 & 1 & 0 & -\sin\psi\, r_y \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -0.2389 \\ 0 & 1 & 0 & -0.1379 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\frac{\partial r_y}{\partial \theta_1} = 0.1819, \quad \frac{\partial r_y}{\partial \theta_2} = -0.1000$$

$$\frac{\partial r_z}{\partial \theta_1} = 0.2758, \quad \frac{\partial r_z}{\partial \theta_2} = 0.1732$$

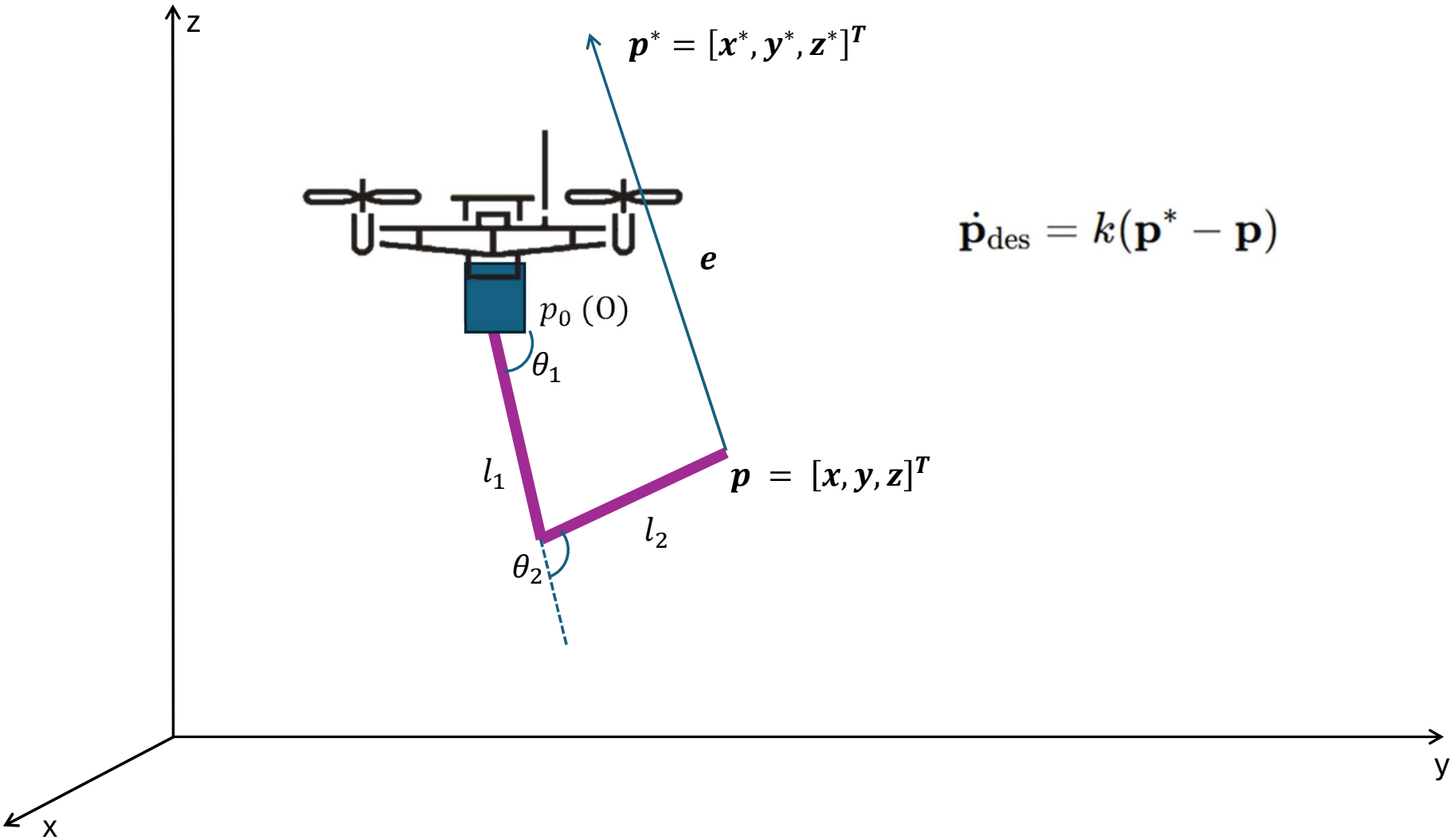
$$J_{\text{arm}} = \begin{bmatrix} -\sin\psi \frac{\partial r_y}{\partial \theta_1} & -\sin\psi \frac{\partial r_y}{\partial \theta_2} \\ \cos\psi \frac{\partial r_y}{\partial \theta_1} & \cos\psi \frac{\partial r_y}{\partial \theta_2} \\ \frac{\partial r_z}{\partial \theta_1} & \frac{\partial r_z}{\partial \theta_2} \end{bmatrix} = \begin{bmatrix} -0.0910 & 0.0500 \\ 0.1575 & -0.0866 \\ 0.2758 & 0.1732 \end{bmatrix}$$

$$\boxed{J_{\text{mobile}} = \begin{bmatrix} 1 & 0 & 0 & -0.2389 & -0.0910 & 0.0500 \\ 0 & 1 & 0 & -0.1379 & 0.1575 & -0.0866 \\ 0 & 0 & 1 & 0 & 0.2758 & 0.1732 \end{bmatrix}}$$

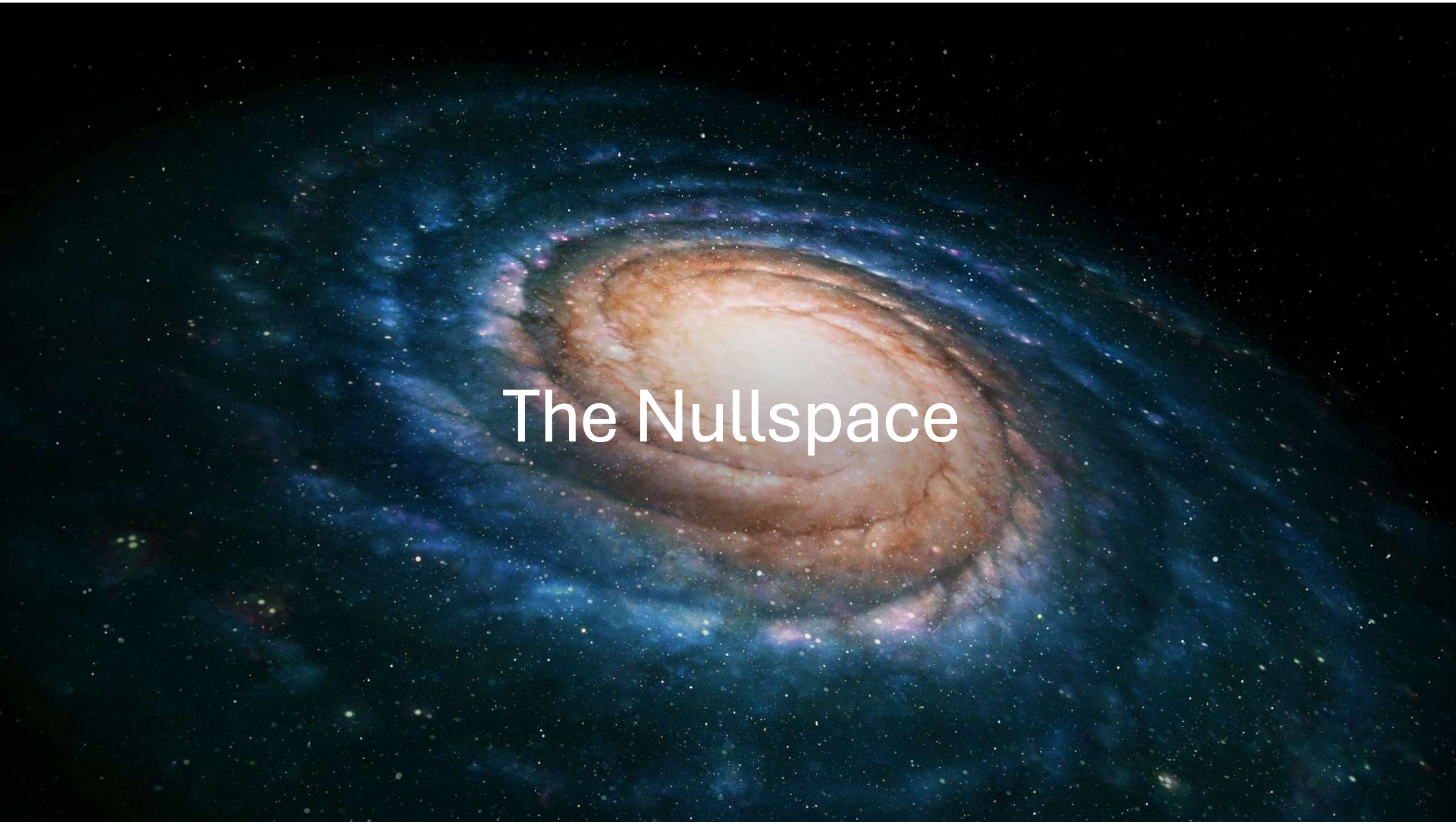
$$\dot{\mathbf{q}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\psi} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = J^{+}_{\text{mobile}}\, \dot{\mathbf{p}}_{\text{des}}, \qquad J^{+} = J^{T}(JJ^{T})^{-1}$$

$$\dot{\mathbf{q}} \approx \begin{bmatrix} 0.3951 \\ 0.3313 \\ -0.0602 \\ -0.1401 \\ -0.0003 \\ -0.0194 \end{bmatrix}$$

Units: $\dot{x}, \dot{y}, \dot{z}$ in m/s, $\dot{\psi}, \dot{\theta}_1, \dot{\theta}_2$ in rad/s.



$$\dot{\mathbf{p}}_{\text{des}} = k(\mathbf{p}^* - \mathbf{p})$$


The Nullspace

We have

$$\dot{\mathbf{p}} = J_{\text{mobile}}\,\dot{\mathbf{q}}$$

$$\dot{\mathbf{p}} \in \mathbb{R}^3 \quad \text{but} \quad \dot{\mathbf{q}} \in \mathbb{R}^6$$

That means many different $\dot{\mathbf{q}}$ can produce the same end-effector motion $\dot{\mathbf{p}}$

6 controls -> 3 task dimensions -> extra freedom

A nullspace motion is a motion of the drone/arm that does not move the end effector (locally)

$$J_{\text{mobile}}\,\dot{\mathbf{q}}_{\text{null}} = 0$$

Key Idea:

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}_{\text{task}} + \dot{\mathbf{q}}_{\text{null}}$$

$$\dot{\mathbf{q}}_{\text{task}} = J^{+}_{\text{mobile}}\,\dot{\mathbf{p}}_{\text{des}}$$

“Same end-effector motion, different robot posture.”

$$\dot{\mathbf{q}}_{\text{task}} = J^{+}_{\text{mobile}}\, \dot{\mathbf{p}}_{\text{des}}$$

$$\dot{\mathbf{q}}_{\text{null}} = \left(I - J^{+}_{\text{mobile}} J_{\text{mobile}}\right) \mathbf{v}$$

$$\boxed{\dot{\mathbf{q}} = J^{+}_{\text{mobile}}\, \dot{\mathbf{p}}_{\text{des}} + \left(I - J^{+}_{\text{mobile}} J_{\text{mobile}}\right) \mathbf{v}}$$

$$\dot{\mathbf{q}}_{\text{task}} = J^{+}_{\text{mobile}}\,\dot{\mathbf{p}}_{\text{des}}$$

$$\dot{\mathbf{q}}_{\text{null}} = \left(I - J^{+}_{\text{mobile}} J_{\text{mobile}}\right)\mathbf{v}$$

$$\boxed{\dot{\mathbf{q}} = J^{+}_{\text{mobile}}\,\dot{\mathbf{p}}_{\text{des}} + \left(I - J^{+}_{\text{mobile}} J_{\text{mobile}}\right)\mathbf{v}}$$

$$\dot{\mathbf{q}}_{\text{task}} = J^{+}_{\text{mobile}}\,\dot{\mathbf{p}}_{\text{des}} \qquad \text{(does the end-effector task)}$$

$$\dot{\mathbf{q}}_{\text{null}} = (I - J^{+}_{\text{mobile}} J_{\text{mobile}})\mathbf{v} \qquad \text{(uses leftover freedom)}$$

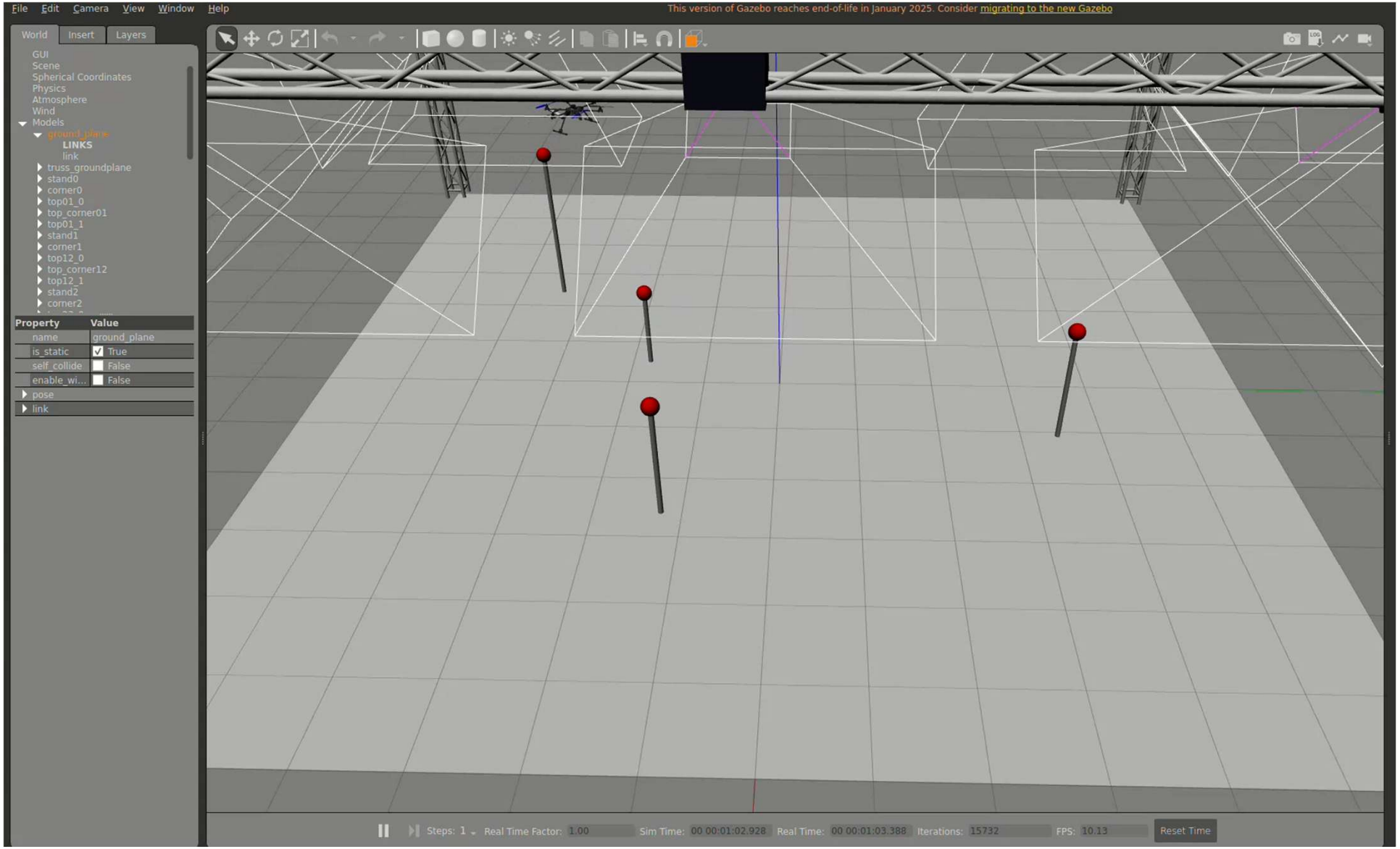
Example: Keep a v shape for the arm

$$\theta_1^* = -45^\circ, \qquad \theta_2^* = 90^\circ$$

$$\mathbf{e}_\theta = \begin{bmatrix} \theta_1^* - \theta_1 \\ \theta_2^* - \theta_2 \end{bmatrix}$$

$$\dot{\mathbf{q}} = J_{\text{mobile}}^{+} \dot{\mathbf{p}}_{\text{des}} + \left(I - J_{\text{mobile}}^{+} J_{\text{mobile}}\right) \mathbf{v} \qquad \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ k_n(\theta_1^* - \theta_1) \\ k_n(\theta_2^* - \theta_2) \end{bmatrix}$$

Goal today:


File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
LINKS
link
truss_groundplane
stand0
corner0
top01_0
top_corner01
top01_1
stand1
corner1
top12_0
top_corner12
top12_1
stand2
corner2
Property Value
name ground_plane
is_static True
self_collide False
enable_wi... False
pose
link
Steps: 1 Real Time Factor: 1.00 Sim Time: 00 00:01:02.928 Real Time: 00 00:01:03.388 Iterations: 15732 FPS: 10.13 Reset Time

Tasks (till end of lecture and until next lecture):

- Add change to launch from Atlassian (We'll talk about git next time)
- Make and test a gimbal node (ROS2, python)
  - Relevant topics: /mavros/imu/data
    /gimbal_controller/commands
- Implement mobile Jacobian for the SDU UAS drone simulator
- Relevant topics: /mavros/setpoint_position/local
  /mavros/state/mode
  /arm_controller/commands
  /end_effector/pose –> check if present; if not, read first point
  /target 1/pose
  /target2/pose etc.
- Try with and without Nullspace.

FLIGHT PROCEDRE:

- Start the Drone in Qgroundcontrol. (wait till it hovers)
- Start your Mobile Jacobian node.
- Wait in the node for Offboard mode to be activated. (/mavros/state/mode)
  OFFBOARD MODE SHALL NEVER BE ACTIVATED FROM A ROS2 NODE. Like..... EVER!
- Move end effector exactly 20cm ABOVE target 1;
- stay there for 5 sek (think about a margin of error);
- move to next target etc.